

RMIT University

School of Engineering

AERO2687 – Intro to Aerospace Engineering

Final Report

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Group: Team Project 49

Thursday 4:30pm - 5:30pm

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Contribution Table

Student Name	Student Number	Contribution
Hugh Williams	[REDACTED]	Support Testing
[REDACTED]	[REDACTED]	Executive Summary, Problem Statement, Wing Design
[REDACTED]	[REDACTED]	Manufacturing Methodology, Manufactured Wing: Measurement and Quality Assurance, conclusion
[REDACTED]	[REDACTED]	Performance test

Group Management

Our group primarily communicated through an Instagram group chat and a dedicated Teams folder and chat. These platforms were used to upload pictures of completed work and to share discussions. Weekly meetings were held during our workshop classes, where we sat down together to plan, problem-solve, and refine our designs.

Tasks were not formally assigned to individuals (except for the specific ones mentioned above). Instead, most activities, such as lay-up of flat panels, foam work, designing the wing core and ply arrangement, and building the wing core, were completed collaboratively. This approach allowed us to continuously discuss what was working, what wasn't, and how to improve as we progressed.

Decision-making was also a group process. Each member contributed their ideas and reasoning, and as a team, we selected the best option. Often this meant combining elements from multiple suggestions to arrive at the most effective solution.

Other Project Management Aspects

The risks associated with this project can be identified from the risk assessment on Canvas. This outlines the potential hazards that we must be aware of during wing box creation.

Other than the risk assessment, hazards can be identified from warning labels found on substances such as epoxy, which we will use. Additionally, from the risk assessment, the hazards we need to be aware of are environmental conditions, hazardous substances, electrical, mechanical, temperature, sound/ noise, and psychological. To mitigate the consequences of activities, we can complete safety training, wear PPE such as glasses, gloves and a lab coat, complete the activity in a correct environment, etc.

When managing our resource allocations, we must be mindful of our limited materials when creating the wing box. We cannot make the air foil too large, as we only have 4 large and 4 small strips of carbon fibre. If the air foil is too large, then the composite layout will be weaker as the sheets are spread out over a greater area. We also have limited access to tools, having only those that can be found in the RMIT workshop and which we are familiar with. This limits our accuracy when creating the air foil to be similar to our chosen air foil. This can also result in inaccuracies when making calculations, as the figures associated with a procreated air foil, such as NACA designs, will differ from reality.

Executive summary

This report demonstrates the design, manufacture, and testing of a small-scale aircraft wing section developed to explore the relationship between aerodynamic performance, structural strength, and manufacturing practicality. The project integrated theoretical design methods, material testing, and experimental evaluation to produce a lightweight, high-strength wing using foam and carbon-fibre composite materials.

Several NACA air foils were analysed to determine the most suitable aerodynamic configuration. The NACA 2412 was selected for its strong lift performance, moderate drag, and forgiving stall behaviour, as well as its compatibility with simple construction techniques. The wing was manufactured using precision-cut foam cores wrapped in carbon-fibre and epoxy resin, ensuring rigidity while manufacturing low mass. The assembly process emphasised accurate alignment, surface finishing, and curing control to preserve aerodynamics shape and strength.

Analytical calculations at 30 m/s predicted a lift of 59.5 N and drag of 11.9 N, yielding a lift-to-drag ratio of 5. Experimental aerodynamic and structural tests confirmed these results within the acceptable margins of error, validating the accuracy of the design and build.

The outcomes demonstrate that the NACA 2412 air foil provides an excellent compromise between aerodynamic efficiency, structural capability, and manufacturability, fulfilling all project objectives and illustrating the successful integration of design theory with hands-on engineering practice.

Introduction/statement of the problem

Designing an aircraft wing requires combining aerodynamic efficiency, structural strength, and manufacturability into a single functional design. Each aspect influences the others: aerodynamic performance determines lift and drag; structural integrity ensures the wing can sustain loads without failure; and manufacturing constraints define what can realistically be built. Understanding and balancing these factors is fundamental to successful aerospace design.

In this project, students are tasked with developing a small-scale wing section that demonstrates these core engineering principles. The design must be aerodynamically efficient, structurally sound, and manufacturable using material such as foam and carbon-fibre reinforcement. This practical exercise connects theoretical analysis with real-world fabrication and testing, bridging the gap between aerodynamic design and structural performance.

The engineering problem addressed in this project is to design a wing structure that achieves high lift-to-drag ratio while maintaining sufficient strength and minimal weight, using simple material and manual manufacturing methods. The selected air foil and internal structure must therefore represent an effective compromise between aerodynamic performance, structural reliability, and ease of construction.

This report outlines the design process, from preliminary testing and air foil evaluation through to manufacturing, aerodynamic assessment, and structural testing of the completed wing.

1. Supporting tests

1.1 - Flat Panel Testing

Three-point bend tests were done on several sizes of flat foam panels as well as five different sandwich composites. When testing the foam thickness, for each 10 mm increase in height, the mass increases by 5 g and the max load increases as well. Three heights were tested, 10 mm, 20 mm, and 30 mm, their max loads were 19.4 N, 72.6 N and 161.2 N respectively. This outlines a clear increasing trend in the power to weight ratio as they are calculated to be 494, 822 and 1174 respectively. From this we can see that having a greater thickness in our wing box will help improve the power to weight ratio.

The five specimens used for the sandwich testing were a thin sandwich with a rib, a wide sandwich with a rib, a thick 9 ply plate, a thin 2 ply plate, and EPS foam. These all had a weight of 35g, 132g, 46g, 10g, and 4g respectively as well as a max load of 221.2N, 708.3N, 585.7N, unidentified, and 15.5N respectively. Excluding the thin 2 ply plate where the max load was not reached, the sandwich with the greatest power to weight ratio was the thick 9 ply plate with 1298. Although its maximum load wasn't the highest, which was the wide sandwich with a rib with 708.3 N max load, its far smaller mass means it is more suitable use in an air foil construction.

To summarise these panel tests, we can infer that the thin 2 ply plate is the most applicable design for our air foil. Its maximum load was reached during testing suggesting it is the strongest structure at the lightest weight. When observing the most optimal height of foam, we can see a non-linear increase in the power to weight ratio relative to the max height. This suggests to us that during the 3-point-bend testing, we want to have the greatest height in foam available to us. This in theory will most improve the air foils maximum load. Although the air foil will need to decrease in thickness from the leading to trailing edge as to generate lift, having the foam at the leading edge be as thick as possible will best impact the power to weight ratio.

1.2 - Flow Visualisation

During our preparation for final tests in the week 10 workshop, we tested the air foil in a flow visualisation wind tunnel. For purposes of accuracy, we should note that the airspeed in the following flow visualisation testing is slower than what will be used in the final testing. From the lift equation, $L = \frac{1}{2} \rho V^2 C_L A$, we can see that a decrease in the airspeed reduces the lift quadratically.

The lower momentum means that the air is less able to stay attached to the air foil's surfaces which can lead to an earlier boundary separation. In *figure 1*, we are shown the air wrapping around the air foil sitting at a 3° angle of attack. This suggests that when testing at faster airspeeds, the air will be better attached to the surface of their air foil as the faster speeds mean it has more kinetic energy. Additionally, the faster flow will increase the lift generated by the air foil as can be seen from the lift equation which was noted above. So, from this initial test we can assume that during the testing phase, our air foil will produce a positive lift to drag ratio.

During this testing, we also tested the air foil at a high angle of attack, seen in *figure 2*. During this section of the testing, we can see the airflow splitting away from the upper surface when meeting the leading edge. This suggests to us that the air foil at a large AOA would not produce any lift and may end up stalling. Whilst this has no impact on the testing phase, it demonstrates the limitations of the air foil's shape when it comes to creating stable and attached airflow at extreme angles of attack needed for efficient lift production.



Figure 1

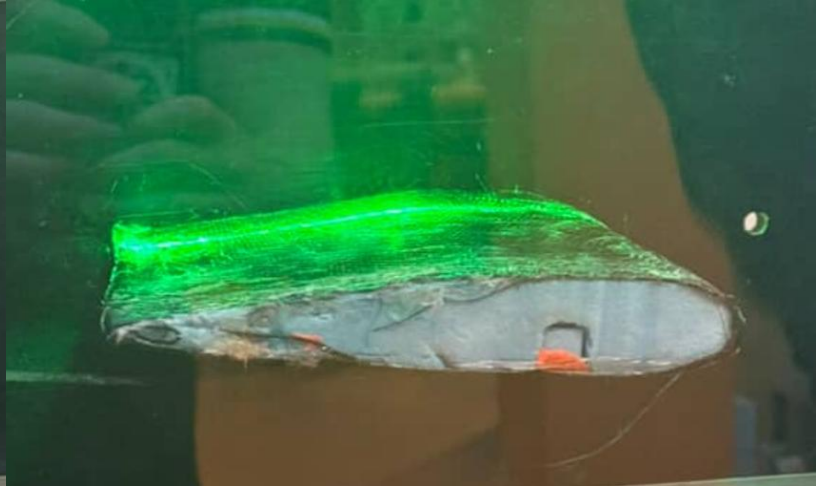


Figure 2

2. Wing Design

2.1 - Air foil Selection and Justification

NACA 23012 (1935):



Figure 3

NACA 23012 1935

The NACA 23012 (1935) (*figure 3*) air foil is being considered for this project due to its balanced aerodynamic performance, structural suitability, and manufacturing simplicity.

Aerodynamically, it provides a strong lift-to-drag ratio with predictable stall

characteristics, making it well suited for stable flight and controllable handling. Its moderate camber ensures efficient lift generation without excessive drag, while the smooth pressure distribution along the chord helps delay flow separation and maintain aerodynamic efficiency.

Structurally, the 12 percent thickness represents an effective compromise between strength and aerodynamic performance. This allows adequate internal volume for spars or mounting points while keeping overall weight low. The rounded leading edge distributes aerodynamic loads smoothly across the surface, reducing stress concentrations, whereas the thin trailing edge- though beneficial for reducing drag, may require reinforcement to prevent structural weakness or vibration.

In terms of manufacturing, the NACA 23012's simple curvature makes it easy to fabricate using foam cores or composite layups. Reinforcement with carbon fibre can further enhance stiffness without significant mass gain.

Overall, the NACA 23012 offers a well-balanced option that aligns with the project's aerodynamic goals, structural requirements, and ease-of-fabrication objectives, making it a strong candidate for further evaluation.

NACA 2412 (1933):

The NACA 2412 (1933) (*figure 4*) air foil is under consideration for this project due to its well-



Figure 4

NACA 2412 1933

rounded aerodynamic performance and practical design characteristics. It's recognised for producing high lift at moderate angles of attack while maintaining smooth, predictable behaviour near stall. The air foil's modest camber and

12% thickness-to-cord ratio contributes to efficient airflow with reduced drag, allowing for stable and responsive handling. These qualities make it suitable for general-purpose applications where both performance and reliability are priorities.

From a structural standpoint, the 12% profile provides ample internal space for spars or reinforcements without excessive weight penalty. The curved leading edge helps to distribute aerodynamic loads gradually, enhancing durability and fatigue resistance, while the tapered trailing edge, although efficient, may need additional material or carbon fibre support to avoid deformation under stress.

In terms of manufacturability, the NACA 2412's smooth and uncomplicated shape lends itself to straightforward fabrication. It can be easily cut from foam, then reinforced with carbon fibre to achieve excellent stiffness-to-weight properties.

Taken together, the NACA 2412 demonstrates a dependable balance of aerodynamic capability, structural robustness, and simplicity of construction, making it a strong contender for the intended design.

NACA 661-212:



Figure 5

NACA 661-212 1940

Developed in 1940, the NACA 661-212 (*figure 5*) represents a more advanced generation of air foil focused on achieving extended laminar flow and exceptionally low drag. Its design maintains smooth airflow over much of the upper and

lower surfaces allowing high aerodynamic efficiency during cruise. The sections low camber and moderate 12% thickness make it ideal for sustained, efficient flight at higher speeds. However, these same features lead to steeper loss of lift and a more abrupt stall when pushed beyond its optimal angle of attack, meaning it requires careful control and clean flow conditions to perform at its best.

Structurally, the 661-212 retains enough depth for internal reinforcement, yet its flatter upper surface changes how loads are distributed along the chord. The sharp trailing edge, while aerodynamically clean, is delicate and typically demands additional strengthening to resist vibration or cracking.

From a manufacturing standpoint, the air foil poses the greatest challenge. Its aerodynamic advantage depends heavily on precision: even small surface imperfections can disrupt laminar flow and increase drag. Achieving this level of accuracy in foam or composite fabrication requires meticulous sanding, alignment, and fibre application.

In essence, the NACA 661-212 offers superior efficiency, but only when manufactured and maintained with high precision.

NACA 23015:



Figure 6

NACA 23015 1935

The NACA 23015 (*figure 6*), introduced in 1935, is a thicker 15% variant of the earlier 23012 section and is being considered for its strong balance between lift

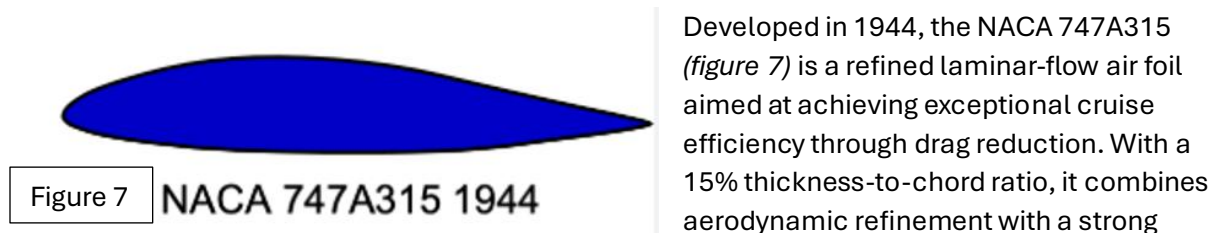
capacity and structural robustness. The increased thickness enhances the air foil's ability to generate lift, particularly at lower speeds, while also providing greater internal volume for spars, reinforcements, or control systems. This thicker profile supports higher load tolerance and greater bending resistance, making it suitable for designs that prioritise durability and strength over minimal drag.

Aerodynamically, the 23015 retains the forgiving stall characteristics typical of the NACA 230 series. The rounded leading edge ensures smooth airflow attachment and gradual stall onset, promoting predictable handling. The trade-off is a modest increase in drag compared to thinner sections, though this is often acceptable in exchange for the improved lift and strength benefits.

Manufacturing the 23015 is straightforward, as its gentle curvature and thicker form are easy to shape accurately from foam. Carbon-fibre wrapping can be applied efficiently to reinforce the surface without significant added weight.

Ultimately, the NACA 23015 provides a sturdy and versatile option, well-suited for applications demanding reliable lift, structural resilience, and ease of fabrication.

NACA 747A315:



Developed in 1944, the NACA 747A315 (figure 7) is a refined laminar-flow air foil aimed at achieving exceptional cruise efficiency through drag reduction. With a 15% thickness-to-chord ratio, it combines aerodynamic refinement with a strong structural core, allowing it to sustain higher loads while maintaining a streamlined shape. The design is optimised to delay the transition from laminar to turbulent flow over a large portion of the surface, delivering impressive performance at steady, moderate angles of attack.

Aerodynamically, the 747A315 offers low drag and good lift characteristics, making it ideal for applications focused on endurance and smooth flight. Its rounded leading edge manages load transfer effectively, promoting stability and consistent pressure distribution. However, the air foil's efficiency is highly sensitive to surface finish- any minor irregularities or roughness can disrupt laminar flow and significantly increase drag.

From a manufacturing perspective, this profile can be shaped accurately using foam cores and carbon-fibre composites. The process, however, demands precision, careful wrapping, and smooth finishing to maintain aerodynamic fidelity.

In summary, the NACA 747A315 rewards precision craftsmanship with superior aerodynamic efficiency, offering excellent cruise performance-though it demands higher build accuracy than conventional or earlier-generation air foils.

Most suitable design - NACA 2412

Following the assessment of multiple air foil configurations - including the NACA 23012, 23015, 661-212, and 747A315 - the NACA 2412 stands out as the most appropriate and balanced choice for this project. Each alternative offered distinct strengths: the NACA 23012 provided dependable aerodynamic performance and predictable stall behaviour; the 23015 delivered greater lift and internal volume but at the cost of increased drag; and the laminar-flow sections (661-212 and 747A315) demonstrated excellent cruise efficiency but were highly sensitive to manufacturing precision and surface imperfections.

The NACA 2412, by contrast, achieves a superior balance across aerodynamic, structural, and manufacturing factors. It offers advanced lift generation, moderate drag, and smooth, stable handling across a wide range of operating conditions. Its 12% thickness-to-cord ratio provides enough space for internal reinforcements without compromising weight or efficiency, while the rounded leading edge supports even load distribution and gentle stall characteristics-important for control and reliability.

From a fabrication perspective, the NACA 2412's simple and well-defined curvature allows easy and repeatable construction using foam cores and carbon-fibre reinforcement. It is less dependent on perfect surface finishes than laminar-flow designs, maintaining good aerodynamic performance even with minor imperfections.

Taken together, these qualities make the NACA 2412 the most suitable and practical air foil application - offering dependable aerodynamic efficiency, structural soundness, and ease of manufacture using available materials and techniques.

3. Manufacturing Methodology

3.1 Materials and Equipment

Carbon Fibre Fabric (150 mm × 250 mm, 4 plies — main wing face sheets/skins)

- **Purpose:** Forms the primary load-bearing skin of the wing, providing high strength and stiffness.
- **Justification:** Carbon fibre has a high strength-to-weight ratio and excellent stiffness, making it ideal for the outer wing surfaces which experience tensile and compressive stresses during loading. The wider plies provide continuous coverage over the main aerodynamic surface, ensuring even stress distribution and minimizing weak joints.

Thin Carbon Fibre Fabric (50 mm × 250 mm, 4 plies — stiffeners/additional layers)

- **Purpose:** Used as localized reinforcements or stiffeners in high-stress areas such as near the root or around the mounting rod.
- **Justification:** The thinner plies allow for precise reinforcement without excessive weight, improving local stiffness. They enable tailored lay-up where extra rigidity is needed without overbuilding the entire structure.

Wind Tunnel Mounting Rod

- **Purpose/ Justification:** Provides the structural interface between the wing and the test rig or wind tunnel mount.

Resin

- **Purpose:** Binds the carbon fibres to the foam, forming a rigid composite once cured.
- **Justification:** Resin will strength the carbon fibre while attaching it to the foam core. It also enables proper load sharing among fibres and creates a smooth aerodynamic surface when cured.

Foam Core (Wing Body/Core Structure)

- **Purpose:** Acts as the lightweight internal core, giving the wing its aerodynamic shape

- **Justification:** The foam provides structural thickness without significant weight. It's easy to machine and ideal for sandwich composite construction.

MDF Blocks (Pressing and Support Tools)

- **Purpose:** Used as support tooling during lay-up and curing to apply even pressure and maintain shape.
- **Justification:** MDF is flat, rigid, making it suitable for applying uniform pressure across the composite surface during curing. This ensures the resin cures evenly and minimizes voids or surface irregularities.

Sandpaper

- **Purpose:** Used to refine and smooth the foam surface before lay-up.
- **Justification:** Smoothing ensures proper contact between the foam and carbon fibre, preventing air gaps and improving adhesion. It also refines the wing's aerodynamic profile to match the intended air foil shape.

Hot Wire Cutter

- **Purpose:** Used to shape the foam into a rough wing profile.
- **Justification:** The hot wire cutter allows for precise, clean cuts along the air foil contour with minimal material removal or dust. It's ideal for producing smooth, accurate geometries.

Pillar Drill

- **Purpose:** Used to accurately form the slot for the metal mounting rod.
- **Justification:** The pillar drill provides controlled, perpendicular drilling, ensuring tight tolerances and proper alignment of the mounting rod. The tool is necessary because the hot wire cutter cannot effectively remove material in confined or circular areas.

Clamps

- **Purpose:** Used to apply uniform pressure to the MDF blocks, holding the assembly securely during curing.
- **Justification:** The clamps ensure consistent compression across the lay-up, maintaining contact between the carbon fibre layers, foam core, and resin. This prevents delamination and promotes proper bonding while the composite cures.

Plastic Bag and Sand

- **Purpose:** To contain the air foil and apply an even distribution of pressure over its surface once the lay-up is complete.
- **Justification:** Uniform pressure is essential to eliminate trapped air bubbles and ensure full contact between the carbon fibre and foam core. Because the air foil has an irregular shape, sand is used inside the plastic bag to conform to the shape of the surface,

providing even pressure when the MDF blocks and clamps are applied. This method helps achieve a smooth finish and strong composite bond throughout the structure.

3.2 Preparation

Safety Equipment/Safety Notes:

- Safety Goggles: Protect eyes from resin splashes or airborne particles during mixing and lay-up.
- Gloves: Prevent skin contact with uncured resin, which can cause irritation, burns, or allergic reactions.
- Lab Coat: Protects clothing and exposed skin from chemical spills and resin contact.
- Hair Tied Back: Prevents hair from becoming contaminated with resin or caught in equipment.
- Ventilation / Respiratory Protection: Resin systems release toxic fumes during mixing and curing. Ensure adequate ventilation or use a room air respirator/fume extraction system to prevent inhalation of harmful vapours.
- Closed Footwear: Shoes must fully cover the feet to protect against spills and dropped tools.
- Supervision: Never work alone in the laboratory. Always ensure a qualified person or instructor is present.
- Hearing and Awareness: Do not wear headphones or any device that limits situational awareness.
- Training Requirement: Only operate equipment for which you have received proper training and approval. Ask for assistance if uncertain.
- Authorised Work Only: Perform only experiments or procedures approved by the instructor. Follow all verbal and written instructions carefully. Unauthorised experiments are strictly prohibited.

Reference [3]

Surface Preparation/ Clean up:

- Cover the workbench with a protective plastic sheet before beginning the lay-up process.
- Ensure the air respirator system is active before handling or mixing resin.
- Keep the workspace free of old or used materials to prevent contamination.
- Use a new mixing cup for each batch of resin. Do not reuse cups.
- After completing the lay-up, remove the plastic cover and thoroughly wipe down the bench with bacterial wipes or other approved cleaning materials.
- Dispose of all used materials and waste (resin cups, gloves, wipes, and off cuts) in the designated hazardous waste bin.
- Leave the workspace clean and ready for the next user

3.3 - Wing Cutting Process

1. Cut Out Foam Block

- Begin by cutting the foam block to the approximate dimensions required for the wing section.
- 2. Mark the Air foil Shape**
 - Place a 1:1 scale cutout of the air foil profile on the side of the foam block.
 - Using a pen or fine marker, trace the perimeter of the air foil shape onto the foam surface.
 - 3. Set Up the Hot Wire Cutter**
 - Ensure the hot wire is properly tensioned and securely fixed in place.
 - Turn on the compressed air nozzle, ensuring it is directed toward the cutting area
 - Power on the hot wire cutter and allow it to reach operating temperature.
 - 4. Rough Cut the Air foil Shape**
 - Using the hot wire cutter, remove excess foam, keeping a small margin between the cutting path and the traced air foil outline to allow for fine sanding later.
 - Once the rough cut is complete, turn off both the wire cutter and compressed air nozzle.
 - 5. Fine Sanding of the Air foil**
 - Move to the sanding station and put on safety goggles.
 - Turn on the compressed air nozzle.
 - Carefully sand the foam surface to the exact air foil perimeter, ensuring smooth and even surfaces.
 - When finished, turn off the compressed air and remove all foam dust from the workstation.
 - 6. Prepare the Mounting Rod Section**
 - On the foam surface, mark and outline the section where the metal mounting rod will be inserted.
 - Put on safety goggles and ensure a supervising instructor is present before operating any power tools.
 - 7. Drill the Mounting Rod Slot**
 - Secure the foam block on the pillar drill table using a machine clamp to prevent movement.
 - Align the marked rod section precisely beneath the drill bit.
 - Switch on the drill and lower it slowly into the foam until reaching the measured depth for the metal rod.
 - Once complete, turn off the drill and vacuum away foam debris.
 - 8. Final Fit and Finishing**
 - Return to the sanding station and smooth the interior of the drilled cavity using sandpaper until the metal rod fits securely in place.
 - After confirming the fit, turn off the compressed air and clean the workstation.
 - Insert the metal rod carefully into position and gently tap it into place to ensure a snug, aligned fit.

3.4 - Lay-Up Process

1. Preparation

- Cover the workbench with a protective plastic sheet before beginning the lay-up process.

- Ensure the air respirator system is active before handling or mixing resin.
- Confirm that all safety equipment is worn: safety goggles, lab coat, gloves, enclosed shoes, and hair tied back.
- Ensure a qualified instructor is present before starting.
- Verify that the foam air foil and all six carbon fibre strips are available and correctly cut to size.

2. Resin Mixing

- Member 1 mixes the resin using a 1:1 ratio of each part (resin and hardener).
- Stir thoroughly until the mixture is uniform and free of air bubbles.

3. Applying Resin and First Carbon Fibre Layer (Front)

- Member 2 uses a syringe to draw approximately 12 mL of resin.
- Apply 6mL of resin evenly onto the first 150 mm × 250 mm carbon fibre fabric.
- Member 3 spreads the resin evenly using a popsicle stick, ensuring fully and evenly spread.
- Member 4 carefully places the fabric onto the front half of the air foil, smoothing it out to remove any bubbles or wrinkles with hands. (*Refer to Appendix 2 for visual reference.*)

4. Applying Resin and Second Carbon Fibre Layer (Back)

- Member 2 applies 6mL of resin on to the second 150 mm × 250 mm carbon fibre fabric.
- Member 3 spreads the resin evenly.
- Member 4 positions the fabric on the back half of the air foil, smoothing out bubbles and ensuring even contact.

5. Applying Additional Reinforcement Layers (50 mm × 250 mm Strips)

- Using a new resin cup mix a fresh resin batch and draw approximately 12 mL into the syringe.
- Apply 3mL of resin to the first 50 mm × 250 mm carbon fibre strip.
- Member 3 spreads the resin evenly, and Member 4 places it between the two main carbon fibre layers along the centre of the air foil. Ensure there are no bubbles or dry spots.
- Apply any remaining resin from the syringe evenly across the surface of the laid-up carbon fibre.

6. Preparing for the Back Lay-Up

- Place a layer of cling wrap beneath the resin-coated air foil surface, then flip or position it to begin the back side lay-up.
- Repeat Steps 3 to 5 for the back half of the air foil.

7. Applying Final Reinforcement Strips

- Apply 3 mL of resin to the third 50 mm × 250 mm carbon fibre strip.
- Evenly spread the resin, then position it along the front curve of the air foil to reinforce the leading edge.
- Repeat the process for the fourth 50 mm × 250 mm carbon fibre strip, placing it along the trailing edge to reinforce the tail section.
- Apply any remaining resin evenly across all exposed carbon fibre surfaces.

8. Vacuum Bagging and Curing Setup

- Wrap the entire air foil in cling wrap to contain excess resin.
- Place the air foil inside a plastic bag and fill the surrounding space with sand, ensuring full coverage and uniform pressure.
- Seal the bag securely.

- Position two MDF blocks on either side of the bag and use six clamps to apply even pressure across the assembly.
- Tighten the clamps until firm pressure is achieved across the entire surface.

9. Post-Lay-Up Clean-Up

- Once the air foil is clamped and secured, begin the clean-up process.
- Remove and dispose of all used resin cups, gloves, and wipes in the designated hazardous waste bin.
- Remove the protective plastic sheet from the workbench and wipe down all surfaces thoroughly using bactericidal wipes or other approved cleaning materials.
- Ensure the workspace is clean and ready for the next use.

3.5 Assembly and Finishing

Overall, there were no major issues encountered during either the lay-up process or the wing-cutting stage. The team worked efficiently and collaboratively, taking turns to complete each task based on individual strengths and experience.

The only minor issue occurred during the wing-cutting process. Although the hot wire cutter had been properly tightened, it did not cut in a perfectly straight line. As a result, when removing excess foam, slightly more material was removed than intended, reducing the overall height of the air foil by approximately 1 cm. Despite this, subsequent sanding ensured that the air foil maintained its intended aerodynamic shape and surface smoothness.

4. Manufactured Wing: Measurement and Quality Assurance

4.1- Visual and Structural Checks

Once the adhesive and resin were left to cure for a week, the wing was removed from the sealed bag. During removal, it was discovered that a single layer of cling wrap had not provided sufficient containment, allowing some resin to leak out. This caused sand from the pressure bag to stick to both the cling wrap and the surface of the wing. However, after the group carefully removed the sand and excess cling wrap, it was found that no damages related to this incident had occurred, and during removal we didn't damage the cured carbon-fibre layers.

After cleaning, the wing was thoroughly inspected, and it was found that a thermal bubble had formed on the surface. This likely occurred due to trapped air or moisture expanding during the curing process, or from uneven heating within the resin. Such bubbles can result from several causes, including entrapped air during lay-up, outgassing of moisture from the foam core, insufficient vacuum pressure, or inadequate resin wet-out. Visual inspection and tap testing confirmed that the defect was a surface blister rather than a deeper delamination, meaning it that it hadn't damaged or effected the core of the wing. We chose to leave the bubble as it was, due to time constraints and the risk of causing further damage to the outer carbon-fibre layer.

4.2 - Discussion of Manufacturing Accuracy and Improvements

To improve the design in future manufacturing runs and prevent inaccuracies or defects, several refinements could be made to the curing and lay-up process. The wing could be sealed more effectively during curing by using multiple layers of cling wrap and ensuring a more uniform distribution of pressure across the surface. A proper vacuum bagging setup would significantly reduce the risk of trapped air or resin leakage by maintaining consistent vacuum pressure, using bleeder and breather layers to allow air and excess resin to escape, and ensuring all seals are airtight. During resin preparation, mixing should be done slowly to avoid entraining air, and care should be taken to achieve full wet-out of the carbon-fibre layers while removing any visible bubbles or wrinkles. Applying even pressure throughout curing, whether through vacuum bagging, or the controlled sandbag method, would also help prevent thermal or gas bubbles from forming.

5. Performance Tests

5.1 - Aerodynamic test

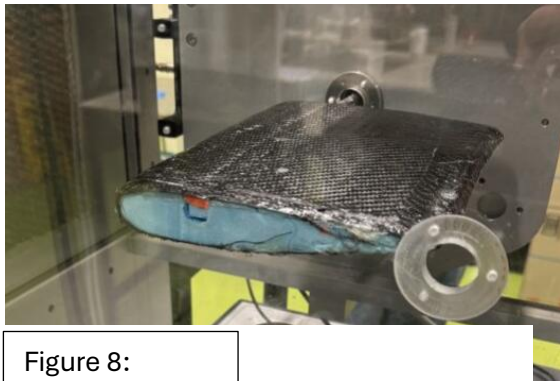


Figure 8:

To evaluate aerodynamics of our wing and compare the validity of our hand calculations of a NACA 2412 [2] aerofoil, we measure the lift-to-drag ratio in a wind tunnel. We attach the wing box using the steel rod we installed during the manufacturing process onto a rod inside the tunnel as seen in *figure 8*.

This rod is mounted to a system that measures the total forces and derives a lift-to-drag ratio from said forces. After 30 seconds of inside the chamber we received a resulting **lift-to-drag ratio of 2.5:0.5 or 5.**

As we refer to our progress report [1] of the hand calculations we processed for a lift-to-drag ratio of a NACA 2412 airfoil [2], we can see that our practical testing and theoretical calculations line up in figure 4:

$$L = C_l \times \frac{1}{2} \rho V^2 A$$

$$L = 0.60 \times \frac{1}{2} \times 1.225 \times 30^2 \times 0.18$$

$$L = 59.535$$

$$D = C_d \times \frac{1}{2} \rho V^2 A$$

$$D = 0.12 \times \frac{1}{2} \times 1.225 \times 30^2 \times 0.18$$

$$D = 11.907$$

$$\frac{L}{D} = \frac{59.535}{11.907} = 5$$

Figure 4

With our results being highly accurate compared to the theoretical calculations, we can see that our manufacturing and replication of the NACA 2412 airfoil is highly accurate. However, this was unexpected due to a slight deformation during the setting of the resin. A shallow, wide bubble had formed along the top of the airfoil which we believe may cause some disadvantages to the lift-to-drag test but clearly did not affect the aerodynamics of the wing largely enough to result in inefficiency.

5.2 - Structural test

To evaluate the total strength-to-weight ratio of our airfoil we first measure the total weight of the wing. On a small scale, the total weight of our wing, including the metal rod, resin, carbon fibre, and structural foam, is 162 grams (*figure 9*).



Figure 9:

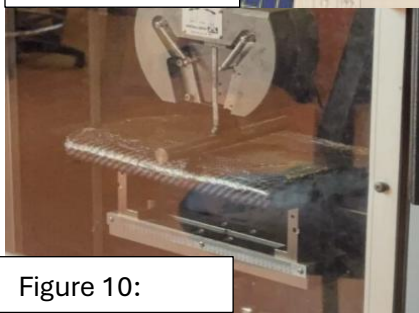


Figure 10:

Now that we've evaluated weight, we now need to measure the strength. The strength test consists of placing the specimen into a hydraulic press with 2 pins on each end and a pin pushing down on the centre of the specimen as seen in *figure 10*. This hydraulic press measures the maximum amount of forced applied before failure of our airfoil.

Now that the specimen is placed, we apply a steadily increasing load until structural failure. The maximum load that the airfoil archived was 1560N. Now that we have got the total weight and the maximum load, we can now calculate the strength-to-weight ratio.

$$S:W = \frac{\text{Maximum Load}(kg)}{\text{Weight}(kg)} = \frac{159.184kg}{0.162kg} = 982.615$$

This means that our airfoil could withstand 983 times its weight.

Throughout the test we experienced a couple of disruptions due to the aforementioned thermal bubble. The load was directly applied on the highest point, which was also the location of the bubble. As the load increased the bubble popped causing a sudden change in deformation and disruption.

The graph shown in *figure 11* shows depict the stress applied compared to the resultant strain of our airfoil, also known as the Young's modulus graph. The sudden spikes and deviations in the data is the press slipping due to the instability of the bubble. The two higher spikes are where the bubbled popped and the press settled on the airfoil.

Overall, these tests allow us to reference our theoretical data into a practical test and verify the validity of our manufacturing processes and the hand calculations. Defects resulted in a reduction in the aerodynamics and structural strength of our airfoil. However, these reductions were minuet and did not heavily affect the final test results.



Figure 11:

Conclusions

This project successfully applied aerodynamic, structural, and manufacturing principles to design and construct a small-scale aircraft wing. The selected NACA 2412 air foil provided an effective balance between lift, drag, and structural performance, while the carbon-fibre and foam composite achieved a strong strength-to-weight ratio.

Minor manufacturing issues occurred, including a small thermal bubble and slight variation in the foam cutting, but these did not significantly affect performance. The project highlighted the importance of even pressure, slow resin mixing, and careful curing control to prevent defects. In future builds, improved sealing methods or vacuum bagging could enhance the finish and accuracy.

Overall, the wing met its performance and design objectives and demonstrated a successful integration of theoretical and practical engineering skills.

References:

[1] [REDACTED], M., Williams, H., [REDACTED] J. and [REDACTED] O. (2025). *AERO2687 - Intro to Aerospace Engineering - Progress Report*.

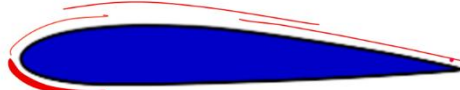
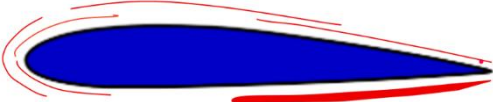
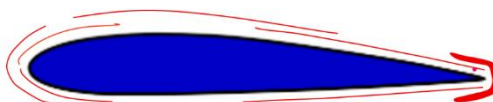
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Appendix

Appendix 1: Progress Report – Team Project 49

Appendix 2: Visual Steps for Wing Lay-Up

Step 3:	1x - 50mm x 250mm  Figure 1: NACA 2412 1933
Step 4:	1x - 150mm x 250mm  Figure 2: NACA 2412 1933
Step 5:	2x - 150mm x 250mm  Figure 3: NACA 2412 1933
Step 6:	2x - 50mm x 250mm  NACA 2412 1933 3x - 150mm x 250mm  NACA 2412 1933  NACA 2412 1933 4x - 150mm x 250mm
Step 7:	3x - 50mm x 250mm  NACA 2412 1933  NACA 2412 1933 4x - 50mm x 250mm

AI Usage Acknowledgment Declaration:

AERO2687 Introduction to Aerospace Engineering

Task: Final Report

Team Number: 49

1. Declaration of AI Use

Please tick the statement that best reflects your team's use of AI tools (e.g., VAL, ChatGPT, Grammarly, Copilot, etc.):

- ☐ We did **not** use any AI tools in preparing this report.
- ☒ We used AI tools **for grammar and language improvements only** (e.g., spelling, punctuation, sentence clarity).
- ☐ We used AI tools to **support** our understanding or generate drafts of technical content, but we **reviewed, verified, and edited** all material for accuracy and appropriateness.
- ☐ We used AI tools in other ways (please explain below).

2. Description of AI Use (if applicable)

Briefly describe how AI tools were used (e.g., summarising technical papers, checking code syntax, formatting tables, refining explanations, generating figures, etc.):

Example: We used ChatGPT to help rephrase our static analysis section for clarity. We cross-checked all content and calculations independently.

We used Grammarly's implemented AI to assess our grammar and improve our language. We double checked these with the spell checker implemented into Microsoft Word Editor which implements the use of Microsoft's AI called 'Copilot'.

3. Verification Statement

We confirm that:

- All technical analysis, data interpretation, and design decisions reflect our own understanding and work.
- Any AI-generated suggestions have been critically evaluated and edited by the team.
- The report meets the academic integrity requirements of this course and the University.

Team Representative Name:

[REDACTED]

Signature:

A handwritten signature in blue ink, consisting of a stylized 'S' followed by a horizontal line.

Date: 19/10/2025